

PHYSICAL CHAIN · EVERY ELEMENT ON THIS DOCUMENT HAS ONE HOME IN IT

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graph LR
    S[System: Linebooker V3 platform, one system] -- "is placed in" --> R[AWS Region: a named AWS place, af-south-1 · ap-southeast-2]
    R -- "hosts one cell" --> C[Cell: one region's complete runtime, L3-01-007 · cell = region in V3]
    C -- "spreads across three zones" --> Z[Availability Zone: one or more data centres, own power, cooling, security]
    Z -- "provides compute and storage" --> I[Compute and storage: where a process actually runs, instance types are L3-12]
  
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AWS REGION AF-SOUTH-1 · AFRICA (CAPE TOWN) · 3 ZONES · OPT-IN

CELL ZA · ONE REGION'S COMPLETE RUNTIME

afs1-az1	afs1-az2	afs1-az3
nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL primary Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007	nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL standby · promoted in cell Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007	nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL standby · read only Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007

Object store · regional, not zonal · stored across all three zones

AWS REGION AP-SOUTHEAST-2 · ASIA PACIFIC (SYDNEY) · 3 ZONES · ON BY DEFAULT

CELL AU · ONE REGION'S COMPLETE RUNTIME

apse2-az1	apse2-az2	apse2-az3
nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL primary Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007	nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL standby · promoted in cell Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007	nats-server 1 of 3 · meta voter Kubernetes nodes node group · 1 of 3 zones Service replicas ≥ 1 replica per service PostgreSQL standby · read only Temporal · 1 of 3 Ingress · 1 of 3 Attached volumes zone bound · do not move open · L4-01-007

Object store · regional, not zonal · stored across all three zones

RUNTIME CHAIN · DRAWN BESIDE THE PLACEMENT, NOT INSIDE IT

```

graph LR
    N[NATS cluster] -- "groups server instances" --> S[Server instance: 3 instances, one per zone]
    S -- "manages replicas" --> D[Service deployment]
    D -- "manages replicas" --> R[Replica: count set by L3-12]
    R --- J[The only join: this instance runs on that compute in that availability zone]
  
```

OVERLAYS · FIVE AXES CROSS, THEY DO NOT NEST (ADR-056, L3-01-010)

A Tenant overlay	B Account overlay	C JetStream domain overlay
business ownership which tenants are served from which cell a tenant may be served from both cells crosses region · L4-01-022	the only machine-enforced wall the NATS account walls data, not availability one account, many cells L3-01-027 · L4-01-023	storage identity T3 and T4 one domain across the supercluster; T5 one domain per cluster za, au and hub in demo 03 · L4-01-024

WORDS ON THIS SHEET	
A Amazon Web Services	meta group the servers that vote on JetStream changes
a availability zone, one or more data centres with its own power	replica one more copy of the same thing
l one region's complete runtime	JetStream domain the storage identity of a cluster
z ion a named AWS place holding several zones	ADR architecture decision record
ne ID the stable zone name, afs1-a21	NATS the messaging system; a product name, not an abbreviation
t-in a region an account must enable before	
T the vote NATS uses to pick a leader	

Cluster Shapes T3, T4 and T5

Which clusters exist, who votes, and what does each shape survive? Defers: which shape is chosen (L4-01-001, open, Architecture); cost per byte, which is sheet 3. **This is a directional document:** a first-pass position as at 18 September 2026. Nothing on this sheet is built.

CLUSTER IS THE AXIS THAT SEPARATES THE THREE SHAPES (L4-01-009)

T3 · GATEWAY SUPERCLUSTER · ONE ARBITER

z

za cluster

3 servers · af-south-1

a

au cluster

3 servers · ap-southeast-2

v

arbiter

1 server · holds no stream assets

gateway · one supercluster

gateway · one supercluster

gateway · one supercluster

7 meta voters · majority 4

one meta group · one JetStream domain

lose one region: 4 voters left, a majority with zero slack

evidence: demo 03 checks C1 to C12

T4 · GATEWAY SUPERCLUSTER · ARBITER CLUSTER OF 3

z

za cluster

3 servers · af-south-1

a

au cluster

3 servers · ap-southeast-2

v

arbiter cluster

3 servers · holds no stream assets

gateway · one supercluster

gateway · one supercluster

gateway · one supercluster

9 meta voters · majority 5

one meta group · one JetStream domain

lose one region: 6 voters left, one node of slack (F11)

survives a region plus one more node, and no further (F12)

evidence: demo 03 checks F1 to F12

T5 · LEAFNODE HUB AND TWO LEAVES

h

hub cluster

3 servers · af-south-1 · domain hub

decision taken 2026-09-18 · L4-01-016

z

za leaf

3 servers · af-south-1 · domain za

a

au leaf

3 servers · ap-southeast-2 · domain au

leafnode Link

leafnode Link

three separate meta groups of 3 · majority 2 each

one JetStream domain per cluster (za, au, hub)

lose the whole hub: both leaves keep a meta leader and still create streams (D10 to D14)

nothing replicates between domains by itself (D15)

evidence: demo 03 checks D1 to D15

COMPARED ON ONE BASIS (L4-01-014)

	T3	T4	T5
Meta voters	7	9	3 + 3 + 3, separate
Majority	4	5	2 in each group
Survives one zone	yes	yes	yes
Survives one region	yes, zero slack	yes, one node of slack	yes, each leaf on its own
When the majority is lost	no new stream or consumer	no new stream or consumer	one group at a time; unaffected groups carry on
Cost to move a byte	gateway traffic crosses the border	same, plus a third region's control traffic	au-to-hub crosses a region; za-to-hub does not

LOSING A MAJORITY BLOCKS CHANGES, NOT TRAFFIC (L4-01-015)

Creating or editing a stream or consumer returns 10008 JetStream system temporarily unavailable	Only after the old leader ages out, about 40 seconds (D03-R5, F7a)	Publishing into a stream that already exists keeps working (F9)
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One axis, three answers. Cluster is the only axis that separates these shapes; placement is the same in all three, and so is every overlay but the JetStream domain. **Built today:** nothing. **Measured:** the voter arithmetic only, as bare nats-server processes on one Mac. **Not shown:** the chosen shape — that is open item L4-01-001 .

LEGEND

Blue boundary

Blue boundary — a gateway supercluster

Red panel

Red panel — a warning or a limit on the evidence

Line label

Line label — names the relationship, never bare "uses"

Purple boundary

Purple boundary — runtime: clusters, instances, deployments, replicas

Blue line

Blue line — a gateway link between clusters

Amber boundary

Amber boundary — cost, rates and rules that govern the sheet

Cyan dotted line

Cyan dotted line — a leafnode link, one thing that crosses a cell boundary

Dashed grey tile

Dashed grey tile — proposed for V3, not built

Arrow head

Arrow head — the direction of the relationship

WORDS ON THIS SHEET

AWS Amazon Web Services

cell one region's complete runtime

RAFT the vote NATS uses to pick a leader

meta group the servers that vote on JetStream changes

quorum / majority more than half the voters

gateway a link joining two NATS clusters into one supercluster

NATS the messaging system; a product name, not an abbreviation

Leafnode a link joining a cluster to a hub, keeping its own meta group

JetStream domain the storage identity of a cluster

arbiter a voter holding no stream data

mirror a copy of one stream in another place

replica one more copy of the same thing

ADR architecture decision record

Traceability: L4-01-009 and L4-01-011 to L4-01-017 and L4-01-030 (cluster axis, the three shapes, the one-basis comparison, quorum-loss behaviour, the hub decision). Evidence: demos/03-multi-cluster-and-accounts/REPORT.md checks C1-C12, F1-F12, D1-D15. Open: L4-01-001 (Architecture), L4-01-005 (Architecture).

LB-V3-L4-01 · 2/3 · CLUSTER SHAPES

Failure Boundaries and Transfer Cost

Which failure is being bought, and what does each shape cost per byte moved? Defers: the actual GB per day per cell (L4-01-004, open) — every figure here is a shape, not a forecast. **This is a directional document:** an initial estimate as at 18 September 2026. Nothing on this sheet is built.

THREE FAILURE BOUNDARIES, TREATED SEPARATELY (L4-01-018)

1 One availability zone

a zone is lost

three servers across three zones keep a majority
answered by all three shapes

CANDIDATE ARBITER HOME

M

ap-southeast-4 · Asia Pacific (Melbourne)

3 zones · opt-in

a second AWS Region in Australia, recorded as a
candidate arbiter home

it is not a cell (L4-01-019)

2 One AWS Region

a whole region is lost

needs voters outside that region
T3 zero slack · T4 one node of slack ·
T5 each leaf on its own

3

One country or jurisdiction

a country is lost or closed

needs a third jurisdiction, not a third region
open · L4-01-003 · business owner

!

Vote-bias analysis

Sydney and Melbourne are separate AWS Regions in ONE country (L4-01-020)

In T3 an arbiter in Melbourne gives Australia 3 + 1 = 4 of 7, a majority.
South Africa holds 3 and can never reach one (L4-01-021)

An arbiter in Melbourne answers an AWS Region failure. It does not answer a country failure.

TRANSFER COST IS A DESIGN INPUT, NOT A FOOTNOTE (L4-01-025)

Rate card · AWS list prices, published 2026-09-16, version 20260916132208

Inside one zone, private addresses **free**
Across a zone, same region
\$0.01 / GB, charged at BOTH ends = \$0.02 / GB
Same rate in af-south-1, ap-southeast-2 and ap-southeast-4

af-south-1 → ap-southeast-2	\$0.147 / GB
ap-southeast-2 → af-south-1	\$0.098 / GB
ap-southeast-4 → af-south-1	\$0.100 / GB
ap-southeast-2 ↔ ap-southeast-4	\$0.080 / GB

cross-region is charged on the SENDING side only; inbound is free

Formula (L4-01-028)

cost per cell per day = D x ((R - 1) + f) x \$0.02

D = GB published per day · R = JetStream replica count
f = share of reads served from a different zone
with R = 3 and f = 0.67 the factor is 2.67

One cross-border GB costs about SEVEN cross-zone GB (\$0.147 against \$0.02)

The zone spread is cheap. The region crossing is the
part worth designing around (L4-01-029).

Worked table · 30-day month

Published per cell per day	Cross-zone, one cell	Mirror it all Cape Town to Sydney	Mirror it all Sydney to Cape Town
1 GB	\$1.60 / mo	\$4.41 / mo	\$2.94 / mo
10 GB	\$16.02 / mo	\$44.10 / mo	\$29.40 / mo
100 GB	\$160.20 / mo	\$441.00 / mo	\$294.00 / mo

excludes committed-use discounts, NAT gateway, Transit Gateway,
PrivateLink and VPC endpoint charges. Every figure depends on
open item L4-01-004.

WHAT EACH SHAPE FORCES ACROSS A BOUNDARY

A

T3 and T4

the arbiter holds no stream assets, so it carries
meta-group control traffic only.
Small and close to constant (L4-01-030).

V

T3, T4 and T5

mirrors and gateway message flow are the variable part.
How much is mirrored is the lever that matters.

H

T5

only au-to-hub bytes cross a region
The hub sits in af-south-1; its placement sets the bill.

Evidence note (L4-01-031)

Demo 03 ran seven topologies as bare nats-server processes on one Mac. It measured quorum and NATS behaviour. It did NOT measure AWS latency, AWS networking, real availability or cost.
Every AWS number on these sheets is a published list price or a published region fact, not a measurement.

Pick the failure first, then read the bill. The failure boundary decides the shape; the shape decides how many bytes cross a border; the rate card turns that into money. **Built today:** nothing. **Measured:** nothing on this sheet — every price is a published list price. **Not shown:** real volumes, which are open item L4-01-004.

SCOPE

One question. Which failure is bought, and what does a byte cost?
Placement is sheet 1, shapes are sheet 2.

DIRECTION

Directional as at 2026-09-18. An initial estimate, not a cost forecast.

SEPARATE

A shape that answers one failure boundary does not answer the
others (L4-01-018).

REGION vs COUNTRY

Sydney and Melbourne are two AWS Regions in one country. Two
regions is not two jurisdictions (L4-01-020).

BIAS

An arbiter in Melbourne biases the vote toward Australia. In T3,
Australia then holds 4 of 7 (L4-01-021).

BOTH ENDS

Cross-zone traffic is charged at both ends. That is why a GB across
a zone costs \$0.02, not \$0.01 (L4-01-026).

ONE WAY

Cross-region traffic is charged on the sending side only. Inbound is
free, so direction changes the bill (L4-01-027).

LEVER

At these volumes the three-zone spread is cheap. The lever is how
much is mirrored, not how many zones are used.

OPEN

Actual GB per day per cell is not known. Every figure here is a
shape, not a forecast (L4-01-004, owned by business and
architecture; L4-01-039).

LEGEND

Green boundary — physical geography: zone, region, country
Amber panel — a rule or a rate that governs the sheet

Amber boundary — cost, rates and rules that govern the sheet
Red panel — a warning or a limit on the evidence

Dashed grey tile — proposed for V3, not built

Dashed amber tile — open, a named owner must confirm

WORDS ON THIS SHEET

AWS Amazon Web Services
AZ availability zone, one or more data centres with
their own power
cell one region's complete runtime
region a named AWS place holding several zones
opt-in a region an account must enable before
use
quorum / **majority** more than half the voters

arbiter a voter holding no stream data
mirror a copy of one stream in another place
replica one more copy of the same thing
GB gigabyte
NATS the messaging system; a product name, not
an abbreviation
NAT network address translation
VPC virtual private cloud, a private AWS network